

PROJECT TECHNICAL REPORT

AI-Powered Penetration Test Report Generator

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1. Project Overview

This project presents an AI-powered penetration testing tool that combines traditional network scanning with modern Large Language Model (LLM) capabilities. The tool automates two critical phases of penetration testing — network reconnaissance and security reporting — by integrating Nmap scanning with the Google Gemini LLM API to generate professional, structured vulnerability reports without manual effort.

The core motivation aligns directly with emerging research in intelligent penetration testing frameworks: leveraging AI to reduce manual overhead, improve accuracy of risk classification, and make security assessments more accessible and reproducible.

2. Objectives

- Automate network and application layer scanning using Nmap to identify open ports, running services, and version information
- Integrate a Large Language Model (LLM) API to intelligently analyze raw scan data and identify vulnerabilities
- Generate structured penetration testing reports with risk classification (Critical / High / Medium / Low)
- Demonstrate practical application of LLMs in cybersecurity automation
- Produce actionable remediation recommendations based on discovered vulnerabilities and CVE references

3. Technical Architecture

The tool is structured as a three-module Python application:

Module	File	Responsibility
Scanner	<code>scanner.py</code>	Executes Nmap scan, parses open ports, services, and versions
Report Generator	<code>report_generator.py</code>	Sends scan data to Gemini LLM API, receives and formats the security report

Main Controller	<code>main.py</code>	Orchestrates the workflow, handles user input, and saves output reports
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4. How It Works — Step by Step

- Step 1: Target Input:** The user provides a target IP address. The tool includes a safety check to ensure only local/private IP ranges are scanned without explicit permission confirmation.
- Step 2: Nmap Network Scan:** The scanner module executes an Nmap scan with service version detection (-sV flag). It identifies all open TCP/UDP ports, the service running on each port, the product name, and the version number — covering both network layer and application layer information.
- Step 3: Data Structuring:** Raw Nmap output is parsed and structured into a JSON object containing host information, open ports, protocols, service names, and version details. This structured format ensures the LLM receives clean, organized data.
- Step 4: LLM Analysis via Gemini API:** The structured scan data is sent to Google Gemini LLM with a carefully crafted prompt instructing it to act as a professional penetration tester. The LLM analyzes the data for known vulnerabilities, CVE references, and security risks.
- Step 5: Report Generation:** Gemini generates a structured security report covering: Executive Summary, Open Ports and Services analysis, Vulnerabilities with CVE references, Risk Classification (Critical/High/Medium/Low), and Prioritized Remediation Recommendations.
- Step 6: Report Saving:** The final report is saved as a timestamped text file in a dedicated reports/ directory for future reference and audit trail purposes.

5. Technologies & Tools Used

Technology	Purpose	Relevance to Security
Python 3.x	Core programming language	Industry standard for security tooling
Nmap	Network port and service scanner	Network/application layer recon — core pentest tool
python-nmap	Python wrapper for Nmap	Programmatic access to scan results
Google Gemini API	Large Language Model (LLM)	Intelligent vulnerability analysis and report writing
JSON	Data structuring format	Clean data transfer between scanner and LLM

6. Sample Report Output Structure

The following illustrates the structure of a report generated by the tool:

```
SECURITY REPORT – TARGET: 127.0.0.1 | DATE: 2026-03-27 1. EXECUTIVE SUMMARY Target system presents MEDIUM overall risk. 3 open ports detected. Two services running
```

```
outdated versions with known CVEs. 2. OPEN PORTS AND SERVICES Port 135 | TCP |  
Windows RPC | Risk: Medium Port 445 | TCP | SMB/CIFS | Risk: High Port 5040 | TCP |  
RPC over HTTP | Risk: Low 3. VULNERABILITIES IDENTIFIED CVE-2017-0144 – EternalBlue  
SMB vulnerability on Port 445 Severity: CRITICAL | Unpatched SMB allows remote code  
execution 4. RISK CLASSIFICATION Critical: 1 finding | High: 1 finding | Medium: 1 |  
Low: 1 5. RECOMMENDATIONS [CRITICAL] Disable SMBv1 immediately – apply MS17-010  
patch [HIGH] Restrict Port 445 access via Windows Firewall rules [MEDIUM] Disable  
RPC endpoint mapper if not required
```

7. Alignment with Intelligent Penetration Testing Research

This project directly addresses the core challenge in modern penetration testing: the manual effort required to interpret raw scan data and produce actionable security reports. By integrating LLMs into the pentest workflow, the tool demonstrates:

- How natural language generation can automate technical report writing in security contexts
- The feasibility of using LLM APIs as intelligent security analysts for network scan interpretation
- A practical framework for combining traditional security tools (Nmap) with AI capabilities
- The potential to extend this approach to web application scanning, IoT device fingerprinting, and automated CVE correlation

These align directly with the research direction of developing Intelligent Penetration Testing Frameworks that leverage AI/ML to enhance and automate the security assessment process.

8. Future Enhancements

- Extend scanning to web application layer using OWASP ZAP or Nikto integration
- Add IoT-specific vulnerability detection for common IoT protocols (MQTT, CoAP, Telnet)
- Implement automated CVE database lookup using NVD API for real-time vulnerability correlation
- Add HTML/PDF report output for professional client-ready deliverables
- Integrate with Metasploit API to suggest exploitation paths based on discovered services

9. Conclusion

This project successfully demonstrates the integration of LLM technology into a penetration testing workflow — producing a functional, end-to-end tool that scans a network target and generates a professional security report automatically. The project bridges traditional cybersecurity techniques with modern AI capabilities, reflecting the growing importance of intelligent automation in security research and practice.